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Week 3 — Training Day 1: Operations Concept Development

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Training Day 1 — Operations Concept Development

Day at a glance

When	What	Output
09:00 – 10:30	Brief: ground-segment elements; ConOps recap; tools intro	Q&A captured
10:30 – 12:00	Mission Planning System (MPS) hands-on	First plan exported
13:00 – 15:00	MCS round-robin: 15-min briefing, then each student rotates through the seven consoles (7 × 15 min)	Position tour completed
15:00 – 16:30	Pass-plan team exercise on the MPS	Team-written pass plan
16:30 – 17:00	Debrief & Day 2 brief; **expected reading flag**	Set up for Day 2

Setting: uOttawa Cyberrange, console room. Simulator runs the **Nominal Post-Commissioning** breakpoint (BP-1 in the breakpoint spec) — the spacecraft is healthy and doing routine ops, so the team can focus on tools rather than recovery.

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1. Brief — what we're flying

EOSAT-1 is a 6U CubeSat (12.0 kg dry, 12.40 kg wet) for ocean-current monitoring along the Canadian coastlines. Its four-band multispectral pushbroom imager (443, 560, 665 and 865 nm) gives a 273 km swath at 33.4 m ground sample distance. It flies a 513 km sun-synchronous orbit at 97.39° inclination with the ascending node at 10:00 local time, so each 94.5-minute orbit includes about 34 minutes of eclipse. Power comes from body-mounted cells and two deployable wings into a 100 Wh battery. Attitude is controlled by four reaction wheels, with two star trackers, six coarse sun sensors, one magnetometer and three gyros, and a 1U green-monopropellant module (66.9 m/s) covers debris avoidance, drag make-up and the end-of-life disposal burn.

The ground segment is five polar S-band stations — Iqaluit, Troll, Svalbard, Inuvik and O'Higgins — on a 5° mask: about 51 passes a day, 7.5 minutes long on average, with a median gap of 33 minutes. Telemetry runs on a three-rung ladder: BEACON 1 kbps, LOW 16 kbps for routine operations, HIGH 128 kbps for image dumps above 10° elevation.

The full brief is textbook Part 2, §5.7. The canonical values are in `configs/eosat1/mission.yaml` and `orbit.yaml`.

1.1 The four ground-segment elements

Textbook Part 2, §8.2, walks the four elements as they appear in the room: **ground stations** (five polar S-band sites on a 5° mask — Iqaluit 63.7° N, Inuvik 68.3° N, Svalbard 78.2° N, Troll 72.0° S and O'Higgins 63.3° S; `ttc.active_gs`, 0x0523, names the live one), **baseband** (the RF and baseband health feed: Eb/N0 and frame sync), the **MCS** (port 9090) and **flight dynamics** (the planner's contact schedule on port 9091 and `tools/pass_gap_analysis.py`). The station list and coordinates are in `orbit.yaml`.

1.2 Pass geometry — the polar S-band picture

Textbook Part 2, §8.2.1, with Figure 8.1 (one orbit with the five stations' visibility cones, lit and eclipsed segments shaded), is the session text. The point to land: the stations sit at high latitude, where the ground tracks of a polar sun-synchronous orbit converge, so most orbits bring the spacecraft over at least one of them. Over 48 h that gives 103 passes, a median gap of 33 minutes and a worst gap of 79 minutes. Passes last 3 to 9.4 minutes; the overhead maximum with the 5° mask is 9.3 minutes at 513 km.

1.3 Procedure structure

Walk through `procedures/leop/first_acquisition.md` using the *Procedure Quick-Reference*, "How to read a procedure file", and textbook Part 2, §7.4. Highlight the header with its entry state, the Prerequisites checklist, steps built from **TC**, **Action**, **Verify** and **Note** labels, Off-Nominal Handling, and the Exit state and Post-Conditions that form the exit criterion. Ownership is not a column in the file: it is `required_positions` in `procedure_index.yaml`. Students read procedures; they do not write them.

1.4 PUS service primer (light)

Use the seven-service list in the *Student MCS & MPS Reference*, §3 (S1, S3, S5, S8, S9, S11 and S17, one line each). The full primer is textbook Part 2, §7.6, and the source is `mcs/pus_services.yaml`. Students command through the MCS; PUS knowledge is context only.

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2. EOSAT-1 mission summary (expected reading before Day 1)

> **Read before Day 1.** Approximate reading time 25 minutes.

The expected reading is textbook Part 2, §5.7 (*What we are flying — EOSAT-1 in 600 words*), with §§5.2–5.5 for the operations concept: the arc of the mission, what an operator must be able to do, the control-room positions, and how passes and shifts work. On the orbit: a sun-synchronous orbit keeps the local solar time of each pass the same, so images of a target are taken under comparable light; EOSAT-1's is at 513 km with the ascending node at 10:00, so it has an eclipse on every orbit. Natural decay from 513 km takes 5.0 years at mean solar activity and 14.2 years at solar minimum, which fails the 5-year disposal rule, so the mission ends with a 34.3 m/s perigee-lowering burn to 390 km (NOM-014), after which re-entry takes 1.5 years (4.2 at solar minimum). The band choice (443, 560, 665 and 865 nm, for dissolved organic carbon and surface chlorophyll) is explained in §5.7. Figures: Figure 8.1 for the orbit and station visibility, and §5.6 for the control-room layout.

Key references

- EOSAT-1 mission config: `configs/eosat1/mission.yaml`
- ECSS-E-ST-70-11C — Space segment operability — https://ecss.nl/standards/active-standards/ecss-e-st-70-11c-space-segment-operability/
- NASA *Space Flight Program and Project Management Requirements* (NPR 7120.5F) — https://nodis3.gsfc.nasa.gov/displayDir.cfm?Internal_ID=N_PR_7120_005F_
- CCSDS Mission Operations Services Concept (CCSDS 520.0-G) — https://public.ccsds.org/Pubs/520x0g4.pdf

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3. Mission Planning System hands-on

Tool: the MPS web UI (port 9091 by default — the planner's own `--port`; the simulator uses 8080 and the MCS 9090).

> **TODO (author):** Write the MPS exercise for this session (about one and a half pages) from `planning/activity_types.yaml`, `planning/imaging_targets.yaml` and `planning/ground_stations.yaml`: open the 24 h station schedule, pick three ocean-current targets, drop one imaging activity per target into a viable window, run the conflict check (slew time, power, thermal), resolve and export the plan, then open it in the MCS Procedures page. Textbook Part 2, §8.5, has a ten-step version that can serve as the base.

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4. MCS round-robin

A 15-minute briefing, then seven 15-minute stations, one per console configured in `mcs/positions.yaml`: the 13:00–15:00 block exactly. Every student visits all seven during the afternoon. At each station, the seat-holder reads the position's role analysis card aloud, then completes one position-specific task while watching live telemetry and records it on the matching workbook card (WB-5.1). Mission Manager and Ground Segment are room roles, not consoles; the ground-segment check is part of the TT&C station.

Position	Role analysis source	15-min task
Flight Director	`mcs/role_analysis/flight_director_role.md`	Run a GO/NO-GO round on the live dashboard (*go for* `CONNECTION_TEST`)
Power & Thermal	`mcs/role_analysis/eps_tcs_role.md`	Predict battery SoC at exit of the next eclipse; name the heater at the highest duty cycle
Flight Dynamics	`mcs/role_analysis/aocs_role.md`	Check `aocs.att_error` against the NOMINAL budget; compute AOS, TCA and LOS for the next Iqaluit pass
TT&C	`mcs/role_analysis/ttc_role.md`	Walk the rate ladder from LOW to HIGH at AOS; confirm which station carries the pass
Payload Operations	`mcs/role_analysis/payload_ops_role.md`	Schedule a single imaging session via MPS; confirm `payload.fpa_ready` = 1
FDIR / Systems	`mcs/role_analysis/fdir_systems_role.md`	Walk through the FDIR panel and explain one rule
Propulsion	`mcs/role_analysis/propulsion_role.md`	Read the arm interlock mask (`prop.arm_inhibit_mask`, 0x0B15) and name the gate blocking an arm, without commanding

The station cards are workbook WB-5.1, one per console, each with its task and the readings to record. The role analysis named in the table is the brief each seat-holder reads aloud.

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5. Pass-plan team exercise

Team produces a written 60–90-min pass plan for an upcoming Iqaluit pass. Plan must include:

- Pass timing (AOS – TCA – LOS, with margins)
- Procedures to run (using the curated procedure quick-reference)
- Imaging window, target ID, predicted ground swath
- Downlink config (data rate, encoding)
- Anomaly contingency: name **two** plausible failures and the procedure ID to run for each

Plans are read aloud at 16:30 and critiqued by another team using the team-axis peer-review worksheet.

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6. Debrief & expected reading for Day 2

> **Expected reading before Day 2.** Read the *Student MCS & MPS Reference* (`student_tool_reference.md`) — sections 1 to 4. Approximate reading time 45 minutes. Bring questions for Day 2 09:00.

Debrief questions:

1. Which MCS position felt most natural to you? Why?
2. What surprised you about a ground-station pass timing-wise?
3. What's one thing you'd want documented better before Day 2?

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7. Tools, files & references

Tools used

- Mission Control System (MCS) — webapp, port 9090
- Mission Planning System (MPS) — webapp, port 9091 (its own server)
- Voice loop — Cyberrange softphone

Simulator files referenced

- `configs/eosat1/mission.yaml`
- `configs/eosat1/orbit.yaml`
- `configs/eosat1/mcs/positions.yaml`
- `configs/eosat1/mcs/role_analysis/*.md`
- `configs/eosat1/planning/*.yaml`
- `configs/eosat1/procedures/procedure_index.yaml`

Open references

- ECSS-E-ST-70C — Ground systems and operations — https://ecss.nl/standards/active-standards/ecss-e-st-70c-ground-systems-and-operations/
- ECSS-E-ST-70-11C — Space segment operability — https://ecss.nl/standards/active-standards/ecss-e-st-70-11c-space-segment-operability/
- CCSDS standards — https://public.ccsds.org/Publications/default.aspx

- NASA Space Flight Program and Project Management Requirements (NPR 7120.5F) — https://nodis3.gsfc.nasa.gov/displayDir.cfm?Internal_ID=N_PR_7120_005F_